

Day 3 Student Handout

Geometry Summer Credit | Transformations, Rotations, and Dilations

Name:	Date:	Period:
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Learning Targets

I can describe translations, reflections, rotations, and dilations on the coordinate plane. I can use rules to find image coordinates and explain how a transformation changes a figure.

Lesson 3A: Translations and Reflections

Warm-Up

1.	Plot A(-3, 2), B(1, 2), and C(-1, 5). Find the slope of AB and the midpoint of AC.
2.	A point moves from P(-4, 1) to P'(2, -3). How many units right/left and up/down did it move?

Concept Launch

Notice and Wonder

A transformation maps every point of a figure to a new point. The original point is usually named A; the image is named A'.

3.	On graph paper, draw triangle ABC with A(-2, 1), B(1, 1), C(0, 4). Translate it 5 units right and 2 units down. Record A', B', and C'.
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Guided Examples

Problem	Work / notes
Translate A(-6, 3) by $\langle 4, -5 \rangle$.	New x: New y: A':
Reflect B(2, -7) across the x-axis.	What changes? B':
Reflect C(-5, 4) across the y-axis.	What changes?

	C':
Triangle has vertices D(-2, -1), E(3, -1), F(1, 2). Reflect across the y-axis.	D': E': F':

Practice

4.	Translate P(3, -4) by $\langle -6, 8 \rangle$.
5.	Reflect Q(-8, -2) across the x-axis.
6.	Reflect R(7, 5) across the y-axis.
7.	A triangle has A(1, 2), B(4, 2), C(2, 6). Translate by $\langle -3, -4 \rangle$. Find A', B', and C'.
8.	Error analysis: A student reflects (-3, 5) across the y-axis and gets (-3, -5). Explain the mistake.

Day 3 Student Handout

Lesson 3B | Rotations and Dilations

Lesson 3B: Rotations and Dilations

Warm-Up

1.	Reflect M(4, -6) across the x-axis and across the y-axis.
2.	Multiply each coordinate by 3: (-2, 5), (0, -4), and (6, 1).

Key Rules

Coordinate Rules About the Origin

Rotate 90 degrees counterclockwise: $(x, y) \rightarrow (-y, x)$. Rotate 180 degrees: $(x, y) \rightarrow (-x, -y)$. Dilate by scale factor k from the origin: $(x, y) \rightarrow (kx, ky)$.

Guided Examples

Problem	Work / notes
Rotate A(3, -2) 90 degrees counterclockwise about the origin.	Use $(x, y) \rightarrow (-y, x)$: A':
Rotate B(-4, 5) 180 degrees about the origin.	Use $(x, y) \rightarrow (-x, -y)$: B':
Dilate C(-3, 6) by scale factor 2 from the origin.	Multiply each coordinate by 2: C':
Dilate triangle P(1, 2), Q(4, 2), R(2, 5) by scale factor $\frac{1}{2}$ from the origin.	P': Q': R':

Practice

3.	Rotate (6, 1) 90 degrees counterclockwise about the origin.
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4. Rotate $(-3, -7)$ 180 degrees about the origin.

5. Dilate $(8, -4)$ by scale factor $\frac{1}{2}$ from the origin.

6. Triangle has vertices $A(-2, 2)$, $B(2, 2)$, $C(0, 6)$. Dilate by scale factor 3 from the origin.

7. Describe what changes and what stays the same when a figure is dilated by scale factor 2.

8. Reflection: Which transformation rule feels most predictable? Which one do you want more practice with?